

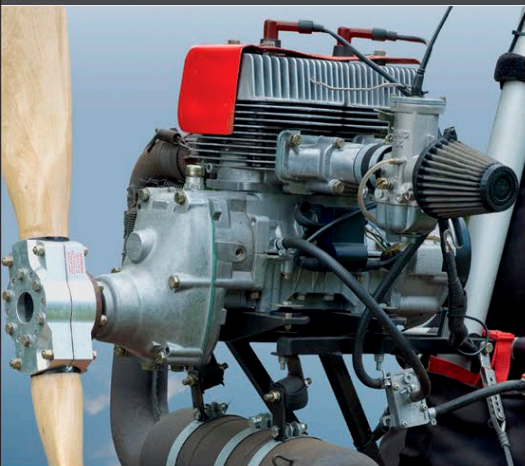
Rotax UL-1V

Rotax is an Austrian company that supplies aircraft engines for ultralights and Light Sport Aircraft (LSAs). Two-stroke engines, such as the Rotax UL-1V, have found a niche in ultralight aircraft, which typically do not fly at high altitudes. The UL-1V is a two-stroke, air-cooled, two-cylinder engine, rated at a respectable 40 hp.

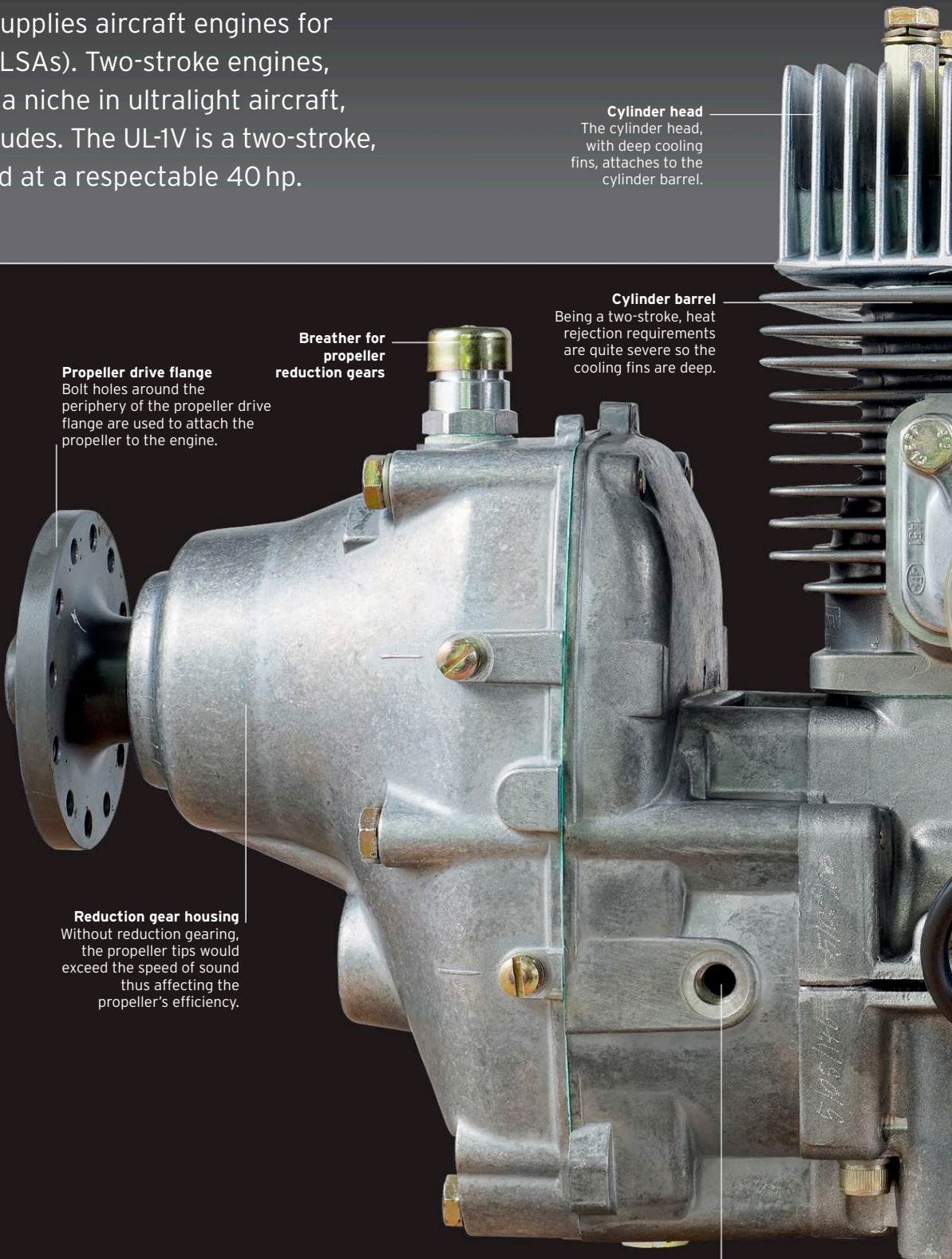
TWO-STROKE MACHINE

Two-stroke engines offer some advantages over their four-stroke counterparts: they are lighter, simpler, and usually far cheaper. However, they also tend to be less reliable, are more sensitive to altitude change with regard to carburetion, and require a more effective cooling system. Four-stroke engines are more commonly used in the aviation industry, but the low cost of ultralight aircraft fitted with two-stroke engines attracts pilots on a limited budget.

ENGINE SPECIFICATIONS	
Dates produced	unknown
Configuration	Air-cooled 2-cylinder 2-stroke inline
Fuel	regulated autofuel
Power output	39.6 hp @ 6,800 rpm
Weight	59 lb (26.8 kg) dry
Displacement	26.64 cu in (0.4366 liter)
Bore and stroke	2.66 in x 2.4 in (67.5 mm x 61 mm)
Compression ratio	9.6:1



► See Piston engines pp.302-303



Propeller drive flange
Bolt holes around the periphery of the propeller drive flange are used to attach the propeller to the engine.

Breather for propeller reduction gears

Reduction gear housing
Without reduction gearing, the propeller tips would exceed the speed of sound thus affecting the propeller's efficiency.

Cylinder head
The cylinder head, with deep cooling fins, attaches to the cylinder barrel.

Cylinder barrel
Being a two-stroke, heat rejection requirements are quite severe so the cooling fins are deep.

Modified for power
Engine cooling may be augmented by an engine-driven fan. Power may be increased by the fitment of two, rather than a single, carburettor.

Engine mount lug
Situating in various parts of the engine are lugs cast integrally to provide mounting locations.